

Repeated Games: Some Definitions

Let $G = (N, \{A_i\}_{i \in N}, \{\pi_i\}_{i \in N})$ be a simultaneous move game and let T be either a positive integer or $+\infty$. A *repeated game of G with time horizon T* (and *publicly observed actions* and *discounting*) is an extensive form game satisfying

1. $H = \cup_{t=0}^T H_t$, where for $t = 0, 1, \dots, T$, H_t is the set of *possible date- t histories*:

$$H_t = \{h_0 a^1 \cdots a^t \mid a^\tau \in A \text{ for } \tau = 1, \dots, t\};$$
2. For $t < T$, each $h_t \in H_t$ begins a *subgame*. In particular, at h_t , each player $i \in N$ *knows* h_{t-1} ;
3. For every $i \in N$, a *utility representation* U_i over the set H_T of terminal histories given by

$$U_i(\tilde{h}_T) = \sum_{t=1}^T \delta^{t-1} \pi_i(\tilde{a}^t),$$

where $\delta \in (0, 1]$ and $\tilde{h}_t = \tilde{h}_{t-1} \tilde{a}^t$ for $t = 1, 2, \dots, T$. If $T = \infty$, then $\delta < 1$. (NB: preferences are assumed to be *additively separable*.)

I often denote a repeated game by $G_T(\delta)$, where G is the stage game, T the time horizon, and $\delta \in (0, 1]$ the discount factor. A *strategy* for $i \in N$ in a repeated game $G_T(\delta)$ is a sequence of functions $s_i = (f_i^1, f_i^2, \dots, f_i^T)$, where $f_i^t : H_{t-1} \rightarrow A_i$ for $t = 1, \dots, T$. Player i 's strategy set S_i is the set of all such sequences of functions. Player i 's preferences over $S = S_1 \times \cdots \times S_n$ are represented by $u_i : S \rightarrow \mathbb{R}$ given by $u_i(s) = U_i(h_T)$, where h_T is the terminal history generated by $s \in S$.

For a stage game G and $i \in N$, define $V_i : A_{-i} \rightarrow \mathbb{R} \cup \{\infty\}$ by

$$V_i(a_{-i}) = \sup_{a_i \in A_i} \pi_i(a_i, a_{-i}).$$

And define v_i to be player i 's *minimax value* in the stage game:¹

$$v_i = \min_{a_{-i} \in A_{-i}} \left\{ \max_{a_i \in A_i} \pi_i(a_i, a_{-i}) \right\}.$$

Of course for any $a^* \in NE(G)$, $u_i(a^*) \geq v_i$. (Why?)

¹If either the “min” or “max” don’t exist, then for some purposes they can be replaced by “inf” or “sup.”